



**PHYSICS
WALLAH**

JEE MAIN 2026

SESSION-01

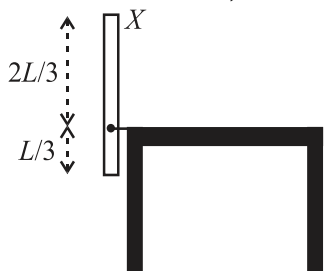
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Shift-02

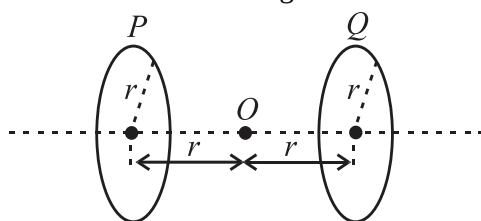
- $$\begin{array}{ll} (1) \quad \frac{3}{5}r & (2) \quad \frac{5}{8}r \\ (3) \quad \frac{4}{5}r & (4) \quad \frac{3}{4}r \end{array}$$

10. A thin uniform rod (X) of mass M and length L is pivoted at a height $\left(\frac{L}{3}\right)$ as shown in the figure.

The rod is allowed to fall from a vertical position and lie horizontally on the table. The angular velocity of this rod when it hits the table top, is (g = gravitational acceleration)



- (1) $\frac{1}{\sqrt{2}}\sqrt{\frac{g}{L}}$ (2) $\sqrt{\frac{3g}{L}}$
 (3) $\frac{3}{\sqrt{2}}\sqrt{\frac{g}{L}}$ (4) $\sqrt{\frac{3g}{2L}}$
11. Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and $4I$ respectively in clockwise direction as seen from O . The net magnetic field at O is:



- (1) $\frac{\mu_0 I}{4\sqrt{2}r}$ towards Q
 (2) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards P
 (3) $\frac{\mu_0 I}{4\sqrt{2}r}$ towards P
 (4) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards Q
12. Five persons P_1, P_2, P_3, P_4 and P_5 recorded object distance (u) and image distance (v) using same convex lens having power $+5D$ as (25, 96), (30, 62), (35, 37), (45, 35) and (50, 32) respectively. Identify correct statement
- (1) Readings recorded by all persons are correct
 (2) Readings recorded by P_4 and P_5 persons are incorrect
 (3) Readings recorded by P_3 person are incorrect
 (4) Readings recorded by P_3 and P_2 persons are incorrect

13. In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division = 0.05 mm, then the least count of the vernier callipers is ____ mm.

- (1) 0.05 (2) 0.002
 (3) 0.02 (4) 0.005

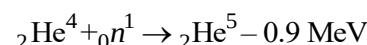
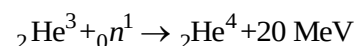
14. Distance between an object and three times magnified real image is 40 cm. The focal length of the mirror used is ____ cm.

- (1) $-15/2$ (2) -20
 (3) -10 (4) -15

15. When a light of a given wavelength falls on a metallic surface the stopping potential for photoelectrons is 3.2 V. If a second light having wavelength twice of first light is used, the stopping potential drops to 0.7 V. The wavelength of first light is ____ m. ($h = 6.63 \times 10^{-34} \text{ J.s}$, $e = 1.6 \times 10^{-19} \text{ C}$, $c = 3 \times 10^8 \text{ m/s}$)

- (1) 3.1×10^{-7} (2) 2.5×10^{-7}
 (3) 2.2×10^{-8} (4) 2.9×10^{-8}

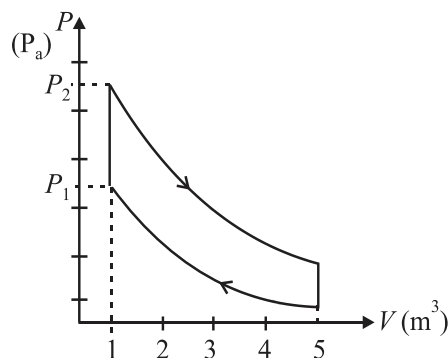
16. The binding energy for the following nuclear reactions are expressed in MeV.



If X_3, X_4, X_5 denote the stability of ${}_2\text{He}^3, {}_2\text{He}^4$ and ${}_2\text{He}^5$, respectively, then the correct order is :

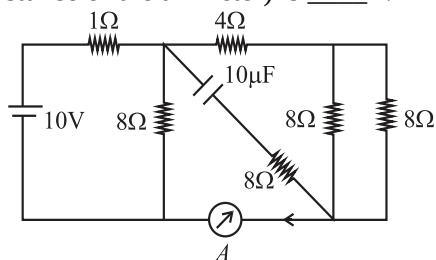
- (1) $X_4 < X_5 < X_3$ (2) $X_4 > X_5 > X_3$
 (3) $X_4 > X_5 < X_3$ (4) $X_4 = X_5 = X_3$

17. 10 mole of an ideal gas is undergoing the process shown in the figure. The heat involved in the process from P_1 to P_2 is α Joule ($P_1 = 21.7 \text{ Pa}$ and $P_2 = 30 \text{ Pa}$, $C_V = 21 \text{ J/K.mol}$, $R = 8.3 \text{ J/mol.K}$). The value of α is



- (1) 21 (2) 24
 (3) 15 (4) 28

18. The reading of the ammeter (A) in steady state in the following circuit (assuming negligible internal resistance of the ammeter) is ____ A.



- (1) 2 (2) 1
(3) 0 (4) $\frac{1}{2}$
19. A cubical block of density $\rho_b = 600 \text{ kg/m}^3$ floats in a liquid of density $\rho_e = 900 \text{ kg/m}^3$. If the height of block is $H = 8.0 \text{ cm}$ then height of the submerged part is ____ cm.
(1) 4.3 (2) 7.3
(3) 5.3 (4) 6.3
20. The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed pipe to that of the open pipe is $5/x$. The value of x is
(1) 4 (2) 1
(3) 3 (4) 2

Integer Type Questions

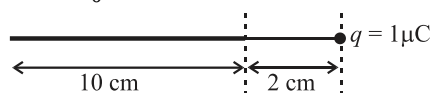
21. A soap bubble of surface tension 0.04 N/m is blown to a diameter of 7 cm . If $(15000 - x) \mu\text{J}$ of work is done in blowing it further to make its diameter 14 cm , then the value of x is $(\pi = \frac{22}{7})$

22. A uniform solid cylinder of length L and radius R has moment of inertia about its axis equal to I_1 . A small co-centric cylinder of length $\frac{L}{2}$ and radius $\frac{R}{3}$ carved from this cylinder has moment of inertia about its axis equals to I_2 . The ratio I_1/I_2 is

23. In a meter bridge experiment to determine the value of unknown resistance, first the resistances 2Ω and 3Ω are connected in the left and right gaps of the bridge and the null point is obtained at a distance $l \text{ cm}$ from the left. Now when an unknown resistance $x\Omega$ is connected in parallel to 3Ω resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is Ω .

24. A point charge $q = 1 \mu\text{C}$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24 \mu\text{C}$, distributed uniformly along its length, as shown in figure. Force between q and wire is ____ N.

(Use : $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$)



25. When 300 J of heat given to an ideal gas with $C_p = \frac{7}{2}R$ its temperature raises from 20°C to 50°C keeping its volume constant. The value of 100 times the ratio of mass of gas to the molecular mass of gas is $(R = 8.314 \text{ J/mol.K})$

SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. At 298 K , the mole percentage of $\text{N}_2(\text{g})$ in air is 80% . Water is in equilibrium with air at a pressure of 10 atm . What is the mole fraction of $\text{N}_2(\text{g})$ in water at 298 K ? (K_H for N_2 is $6.5 \times 10^7 \text{ mm Hg}$)
(1) 1.17×10^{-4} (2) 9.35×10^{-5}
(3) 1.23×10^{-7} (4) 9.35×10^5
27. The wavelength of spectral line obtained in the spectrum of Li^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2, is
(1) $2.28 \times 10^{-7} \text{ cm}$
(2) $1.14 \times 10^{-7} \text{ cm}$
(3) $2.28 \times 10^{-6} \text{ cm}$
(4) $1.14 \times 10^{-6} \text{ cm}$

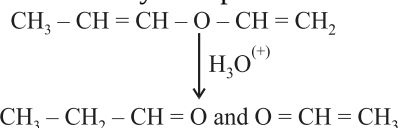
28. The number of possible tripeptides formed involving alanine (ala), glycine (gly) and valine (val), where no amino acid has been used more than once is:

- (1) 8 (2) 6
(3) 4 (4) 3

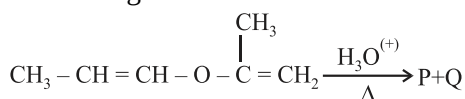
29. The heat of atomisation of methane and ethane are ' x ' kJ mol^{-1} and ' y ' kJ mol^{-1} respectively. The longest wavelength (λ) of light capable of breaking the C-C bond can be expressed in SI unit as:

- (1) $\frac{N_A hc}{250(y-6x)}$ (2) $\frac{N_A hc}{250(4y-6x)}$
(3) $\frac{hc}{1000} \left(\frac{y-6x}{4} \right)^{-1}$ (4) $N_A hc \left(y - \frac{6x}{4} \right)^{-1}$

30. The unsaturated ether on acidic hydrolysis produces carbonyl compounds as shown below:

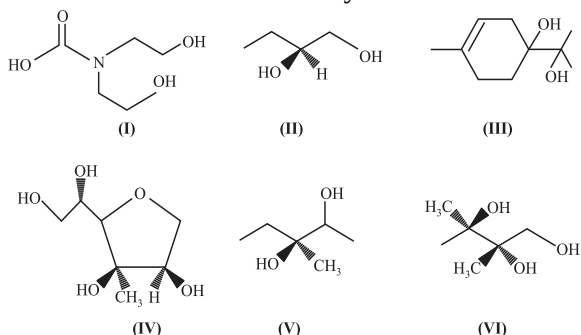


Based on this, predict the solution/reagent that will help to distinguish "P" and "Q" obtained in the following reaction:

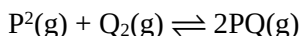


- (1) Fehling solution
(2) Lucas reagent
(3) 2, 4 - DNP reagent
(4) Saturated NaHSO_3 solution
31. One mole of $\text{Cl}_2(\text{g})$ was passed into 2L of cold 2M KOH solution. After the reaction, the concentrations of Cl^- , ClO^- and OH^- are respectively (assume volume remains constant)
- (1) 0.5M, 0.5M, 0.5M
(2) 1M, 1M, 1M
(3) 0.5M, 0.5M, 1M
(4) 0.75M, 0.75M, 1M

32. From the following, how many compounds contain at least one secondary alcohol?



- (1) Two
(2) Three
(3) Five
(4) Four
33. Consider the following gaseous equilibrium in a closed container of volume 'V' at T(K).



2 moles each of $\text{P}_2(\text{g})$, $\text{Q}_2(\text{g})$ and $\text{PQ}(\text{g})$ are present at equilibrium. Now one mole each of ' P_2 ' and ' Q_2 ' are added to the equilibrium keeping the temperature at T(K). The number of moles of P_2 , Q_2 and PQ at the new equilibrium, respectively, are

- (1) 1.66, 1.66, 1.66
(2) 2.67, 2.67, 2.67
(3) 1.21, 2.24, 1.56
(4) 2.56, 1.62, 2.24

34. "X" is an oxoanion of the lightest element of group 7 (in the periodic table). The metal is in +6 oxidation state in "X". The color of the potassium salt of X is

- (1) green (2) yellow
(3) orange (4) purple

35. Two liquids A and B form an ideal solution at temperature T K. At T K, the vapour pressures of pure A and B are 55 and 15 kN m^{-2} respectively. What is the mole fraction of A in solution of A and B in equilibrium with a vapour in which the mole fraction of A is 0.8?

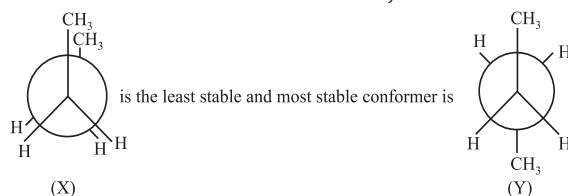
- (1) 0.480 (2) 0.340
(3) 0.5217 (4) 0.663

36. The correct order of C, N, O and F in terms of second ionisation potential is

- (1) $\text{C} < \text{N} < \text{F} < \text{O}$ (2) $\text{C} < \text{O} < \text{N} < \text{F}$
(3) $\text{F} < \text{N} < \text{C} < \text{O}$ (4) $\text{C} < \text{F} < \text{N} < \text{O}$

37. Given below are two statements:

Statement I: There are several conformers for n-butane. Out of those conformers,



Statement II: As the dihedral angle increases, torsional strain decreases from (X) to (Y). In the light of the above statements, choose the correct answer from the options given below

- (1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is false but Statement II is true
(4) Statement I is true but Statement II is false

38. Given below are two statements:

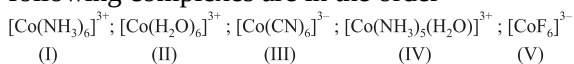
Statement I: Cross aldol condensation between two different aldehydes will always produce four different products.

Statement II: When semicarbazide reacts with a mixture of benzaldehyde and acetophenone under optimum pH, it forms a condensation product with acetophenone only.

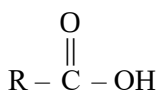
In the light of the above statements, choose the correct answer from the options given below

- (1) Statement I is true but Statement II is false
(2) Both Statement I and Statement II are false
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are true

39. The wavelength of light absorbed for the following complexes are in the order



- (1) III < I < IV < II < V
 (2) III < IV < I < II < V
 (3) III < I < IV < V < II
 (4) III < I < II < IV < V
40. Pair of species among the following having same bond order as well as paramagnetic character will be-
- (1) O_2^- , N_2^+ (2) O_2^- , N_2^-
 (3) O_2^+ , N_2^- (4) O_2^+ , N_2^{2-}
41. Find out the statements which are not true.
- A. Resonating structures with more number of covalent bonds and lesser charge separation are more stable.
 B. In electromeric effect, an unsaturated system shows +E effect with nucleophile and -E effect with electrophile.
 C. Inductive effect is responsible for high melting point, boiling point and dipole moment of polar compounds.
 D. The greater the number of alkyl groups attached to the doubly bonded carbon atoms, higher is the heat of hydrogenation.
 E. Stability of carbanion increases with the increase in s character of the carbon carrying the negative charge. Choose the correct answer from the options given below:
- (1) A, C & D only (2) A, D & E only
 (3) B & D only (4) B, D & E only
42. A student has planned to prepare acetanilide from aniline using acetic anhydride. The student has started from 9.3 g of aniline. However, the student has managed to obtain 11 g of dry acetanilide. The % yield of this reaction is:
- (1) 97.5% (2) 81.5%
 (3) 59.5% (4) 72.5%
43. In the Group analysis of cations, Ba^{2+} & Ca^{2+} are precipitated respectively as
- (1) sulphide & sulphide
 (2) carbonate & carbonate
 (3) chromate & sulphide
 (4) hydroxide & carbonate
44. Given below are two statements:
- Statement I:** The dipole moment of R-CN is greater than R-NC and R-NC can undergo hydrolysis under acidic medium to produce



Statement II: R-CN hydrolyses under acidic medium to produce a compound which on treatment with SOCl_2 , followed by the addition of NH_3 gives another compound(x). This compound (x) on treatment with NaOCl/NaOH gives a product, that on treatment with $\text{CHCl}_3/\text{KOH}/\Delta$ produces R-NC

In the light of the above statements, choose the correct answer from the options given below

- (1) Both Statement I and Statement II are true
 (2) Both Statement I and Statement II are false
 (3) Statement I is false but Statement II is true
 (4) Statement I is true but Statement II is false
45. Choose the INCORRECT statement
- (1) CO_2 is the most acidic oxide among the dioxides of group of 14 elements.
 (2) Among the isotopes of carbon, ^{13}C is a radioactive isotope.
 (3) Carbon cannot exceed its covalency more than four.
 (4) Carbon exhibits negative oxidation states along with +4 and +2.

Integer Type Questions

46. 0.25 g of an organic compound "A" containing carbon, hydrogen and oxygen was analysed using the combustion method. There was an increase in mass of CaCl_2 tube and potash tube at the end of the experiment. The amount was found to be 0.15 g and 0.1837 g, respectively. The percentage of oxygen in compound A is % (Nearest integer).
 (Given: molar mass in g mol^{-1} H: 1, C: 12, O: 16)
47. Grignard reagent RMgBr (P) reacts with water and forms a gas (Q). One gram of Q occupies 1.4 dm^3 at STP. (P) on reaction with dry ice in dry ether followed by H_3O^+ forms a compound (Z). 0.1 mole of (Z) will weigh g. (Nearest integer)
48. The half-life of ^{65}Zn is 245 days. After x days, 75% of original activity remained. The value of x in days is (Nearest integer) (Given: $\log 3 = 0.4771$ and $\log 2 = 0.3010$)
49. Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be $\frac{1}{30}$ of the molar conductivity of another weak acid HZ with concentration of 0.02 M. If $\lambda_{\text{Q}^-}^\circ$ happened to be equal with $\lambda_{\text{Z}^-}^\circ$, then the difference of the pK_a values of the two weak acids ($\text{pK}_a(\text{HQ}) - \text{pK}_a(\text{HZ})$) is (Nearest integer). [Given: degree of dissociation (α) $\ll 1$ for both weak acids, λ° : limiting molar conductivity of ions]

50. A chromium complex with a formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ has a spin only magnetic moment value of 3.87 BM and its solution conductivity corresponds to 1: 2 electrolyte. 2.75 g of the complex solution was initially passed through a cation exchanger. The solution obtained after the

process was reacted with excess of AgNO_3 . The amount of AgCl formed in the above process is g. (Nearest integer)

[Given: Molar mass in g mol^{-1} Cr: 52; Cl: 35.5, Ag:108, O:16, H:1]

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$, and

$$\lim_{t \rightarrow x} \frac{t^2 y(x) - x^2 y(t)}{x - t} = 3 \text{ for each } x > 0. \text{ Then}$$

$2y(2)$ is equal to

- (1) 27 (2) 18
(3) 23 (4) 12

52. Let the length of the latus rectum of an ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b), \text{ be } 30. \text{ If its eccentricity is}$$

the maximum value of the function

$$f(t) = -\frac{3}{4} + 2t - t^2, \text{ then } (a^2 + b^2) \text{ is equal to}$$

- (1) 516 (2) 496
(3) 256 (4) 276

53. Let a_1, a_2, a_3, a_4 be an A.P. of four terms such that each term of the A.P. and its common difference l are integers. If $a_1 + a_2 + a_3 + a_4 = 48$ and $a_1 a_2 a_3 a_4 + l^4 = 361$, then the largest term of the A.P. is equal to

- (1) 24 (2) 27
(3) 23 (4) 21

54. Let $X = \{x \in N: 1 \leq x \leq 19\}$ and for some $a, b \in R, Y = \{ax + b : x \in X\}$. If the mean and variance of the elements of Y are 30 and 750, respectively, then the sum of all possible values of b is

- (1) 100 (2) 20
(3) 80 (4) 60

55. Let $\vec{a} = 2\hat{i} - 5\hat{j} + 5\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 3\hat{k}$. If $\vec{c}$ is a vector such that $2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0}$ and $(\vec{a} - \vec{b}) \cdot \vec{c} = -97$, then $|\vec{c} \times \vec{k}|^2$ is equal to

- (1) 193
(2) 233
(3) 205
(4) 218

56. Let the angles made with the positive x-axis by two straight lines drawn from the point $P(2, 3)$ and meeting the line $x + y = 6$ at a distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2 . Then the value of $(\theta_1 + \theta_2)$ is:

- (1) $\frac{\pi}{2}$ (2) $\frac{\pi}{3}$
(3) $\frac{\pi}{12}$ (4) $\frac{\pi}{6}$

57. Let the image of parabola $x^2 = 4y$, in the line $x - y = 1$ be $(y + a)^2 = b(x - c)$, $a, b, c \in N$. Then $a + b + c$ is equal to

- (1) 6 (2) 8
(3) 4 (4) 12

58. The largest value of n , for which 40^n divides $60!$, is

- (1) 13 (2) 12
(3) 11 (4) 14

59. If the domain of the function $f(x) = \sin^{-1}\left(\frac{1}{x^2 - 2x - 2}\right)$ is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta$ is equal to

- (1) 3 (2) 5
(3) 4 (4) 2

60. Let $[t]$ denote the greatest integer less than or equal to t . If the function

$$f(x) = \begin{cases} b^2 \sin\left(\frac{\pi}{2} \left[\frac{\pi}{2} (\cos x + \sin x) \cos x \right] \right), & x < 0 \\ \frac{\sin x - \frac{1}{2} \sin 2x}{x^3}, & x > 0 \\ a, & x = 0 \end{cases} \text{ is}$$

continuous at $x = 0$, then $a^2 + b^2$ is equal to

- (1) $\frac{9}{16}$ (2) $\frac{1}{2}$
(3) $\frac{5}{8}$ (4) $\frac{3}{4}$

61. Let $\vec{a} = 2\hat{i} - \hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} + 3\hat{k}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|^2$

is equal to

- (1) 13
(2) $\frac{\sqrt{35}}{2}$
(3) $\frac{\sqrt{21}}{2}$
(4) 7

62. The sum of all values of α , for which the shortest distance between the lines $\frac{x+1}{\alpha} = \frac{y-2}{-1} = \frac{z-4}{-\alpha}$

and $\frac{x}{\alpha} = \frac{y-1}{2} = \frac{z-1}{2\alpha}$ is $\sqrt{2}$, is

- (1) 8 (2) 6
(3) -8 (4) -6

63. Let f be a function such that $3f(x) + 2f\left(\frac{m}{19x}\right) = 5x$, $x \neq 0$, where

$m = \sum_{i=1}^9 (i)^2$. Then $f(5) - f(2)$ is equal to

- (1) 18
(2) 36
(3) 9
(4) -9

64. The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is

- (1) 1581
(2) 1578
(3) 1579
(4) 1580

65. Consider the following three statements for the function $f: (0, \infty) \rightarrow \mathbb{R}$ defined by

$$f(x) = \log_e x - |x-1|:$$

- (I) f is differentiable at all $x > 0$.
(II) f is increasing in $(0, 1)$.
(III) f is decreasing in $(1, \infty)$. Then.
(1) All (I), (II) and (III) are TRUE.
(2) Only (I) and (III) are TRUE.
(3) Only (II) and (III) are TRUE.
(4) Only (I) is TRUE.

66. Let $f(x) = \int \frac{7x^{10} + 9x^8}{(1+x^2+2x^9)^2} dx, x > 0$,

$$\lim_{x \rightarrow 0} f(x) = 0 \text{ and } f(1) = \frac{1}{4}.$$

$$\text{If } A = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{4} & f'(1) & 1 \\ \alpha^2 & 4 & 1 \end{bmatrix} \text{ and } B = \text{adj}(\text{adj} A) \text{ be such}$$

that $|B| = 81$, then α^2 is equal to

- (1) 3 (2) 4
(3) 2 (4) 1

67. Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x=0, x=1, y^2=x$ and $y=|\alpha x-5| - |1-\alpha x| + \alpha x^2$.

Then $(f(0) + f(1))$ is equal to

- (1) 7 (2) 12
(3) 14 (4) 9

68. $\left(\frac{1}{3} + \frac{4}{7}\right) + \left(\frac{1}{3^2} + \frac{1}{3} \times \frac{4}{7} + \frac{4^2}{7^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^2} \times \frac{4}{7} + \frac{1}{3} \times \frac{4^2}{7^2} + \frac{4^3}{7^3}\right) + \dots$ upto infinite terms, is equal to

- (1) $\frac{4}{3}$ (2) $\frac{6}{5}$
(3) $\frac{7}{4}$ (4) $\frac{5}{2}$

69. Let $P = [p_{ij}]$ and $Q = [q_{ij}]$ be two square matrices of order 3 such that $q_{ij} = 2^{(i+j-1)} p_{ij}$ and $\det(Q) = 2^{10}$. Then the value of $\det(\text{adj}(\text{adj} P))$ is:

- (1) 81 (2) 32
(3) 124 (4) 16

70. The smallest positive integral value of a , for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to

- (1) 3 (2) 4
(3) 9 (4) 7

Integer Type Questions

71. Let $z = (1+i)(1+2i)(1+3i)\dots(1+ni)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then n is equal to ____

72. Let S be a set of 5 elements and $P(S)$ denote the power set of S . Let E be an event of choosing an ordered pair (A, B) from the set $P(S) \times P(S)$ such that $A \cap B = \emptyset$. If the probability of the event E is $\frac{3^p}{2^q}$, where $p, q \in N$, then $p + q$ is equal to
73. The number of elements in the set $\{x \in [0, 180^\circ]: \tan(x + 100^\circ) = \tan(x + 50^\circ) \tan x \tan(x - 50^\circ)\}$ is

74. If $f(x)$ satisfies the relation $f(x) = e^x + \int_0^1 (y + xe^x) f(y) dy$, then $e + f(0)$ is equal to
75. Let (h, k) lie on the circle $C: x^2 + y^2 = 4$ and the point $(2h + 1, 3k + 2)$ lie on an ellipse with eccentricity e . Then the value of $\frac{5}{e^2}$ is equal to

